Building Seismic-Resilience Credit Model

Poisson Occurrence, Fault-Geometry Spatial Draw, GMPE Intensity and a Lognormal Fragility Chain

MKM Research Labs

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David K Kelly

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Contents

1 Executive Summary

The Building Seismic-Resilience Credit Model (`MKM-SEIS-001`) prices the seismic resilience of a commercial asset so that seismic loss aggregates into the same resilience-credit currency as the existing 10,000-storm Monte Carlo engine and the fire model (`MKM-FIRE-001`). The credit is consumed by the Commercial PRS Pricer as an independent *seismic* leg, combined with the flood, wind and fire legs through a root-sum-of-squares all-in coupon. The model is built from four governance-separated component models:

- **Model A — Occurrence.** A Poisson occurrence model with per-asset annual rate $\lambda_i = \lambda_{\text{zone}}\, m_{\text{fault}}\, m_{\text{soil}}$. A spatial sub-layer draws the rupture location along the catchment fault-trace polyline — structured identically to the river-centreline geometry used in the storm module — and computes the Joyner-Boore source-to-site distance R_{JB}.
- **Model B — Ground Motion Intensity.** A Ground Motion Prediction Equation (GMPE) maps the sampled (M, R_{JB}) pair to a site peak ground acceleration (PGA), adjusted for local soil amplification through V_{S30}.
- **Model C — Seismic Response Effectiveness.** A deterministic mapping from the asset's BRI geoseismic measures (GS01–GS18) to the fragility median multipliers, the soil-amplification gate, the post-earthquake-fire cascade probability and the recovery acceleration factor consumed by Model D.
- **Model D — Damage State, Loss & Resilience Credit.** A single-step lognormal fragility draw assigns one of four damage states (DS0–DS3). Aggregated over n_{sim} draws the model emits annual expected loss, the no-collapse rate, and the full-collapse (DS3) frequency in basis points — the seismic leg the PRS pricer consumes.

The *point-of-no-return analogue* is DS3 (complete damage / collapse): once this state is assigned, loss is 100% and recovery requires demolition and rebuild. The DS3 frequency, $n_{\text{DS3}}/n_{\text{sim}} \times 1.0 \times 10{,}000$, is the bps the PRS pricer charges for the seismic leg.

The model is designed to be *sensitive to all three primary axes*: **distance** (the source-to-site R_{JB} governs GMPE attenuation), **intensity** (the hazard zone governs occurrence frequency and the magnitude distribution) and **resilience** (the BRI geoseismic measures shift the fragility medians). Section 7 demonstrates each.

Status: Development (Tier 2, RAG Amber). All four component models, the end-to-end orchestrator, the port stage, the PRS-pricer and Loan-Calculator integration and the sensitivity harness are delivered with unit tests; the six Section 7 acceptance gates all pass. Every numeric parameter is a first-pass engineering-judgement seed (`config/seismicmatrices.json` and `config/seismic_zones.json`, `version "seed-0"`); the results in this document are therefore *directional, not calibrated.*

Key risk: The model uses placeholder calibration throughout. It is not approved for production until historical calibration completes and the Model Risk Committee (MRC) issues sign-off (remediation items SEIS-R1–R5).

2 Model Purpose and Scope

2.1 Purpose

To assign each commercial asset a seismic-resilience credit — an annual unconditional loss frequency, a conditional no-collapse rate and a full-collapse (DS3) frequency in bps — priced pari-passu with the flood, wind and fire hazards. The credit rewards genuine, evidenced seismic resilience (seismic foundation design, lateral force resistance, base isolation, independent audit) and penalises the configurations that historically drive total losses, most notably unreinforced masonry or pre-code concrete frames in high-hazard zones without base isolation or lateral bracing.

2.2 Scope

The model covers commercial assets described by the v2 commercial-asset and resilience sections of the Common Data Model (CDM). It reads only the derived `AssetSeismicFeatures` bundle — never the raw CDM record — so the feature contract is explicit, testable and stable against CDM schema churn. The in-scope drivers are commercial type, construction type, number of storeys, soil type, seismic hazard zone, the eighteen BRI geoseismic measures (GS01–GS18) and fault proximity. The catchment fault trace is a companion spatial file (`data/catch/<name>/fault_trace.json`) structured identically to the river polyline used in the storm module.

2.3 Out of Scope

- Finite-element or nonlinear time-history structural analysis.
- Casualty, life-safety or evacuation modelling.
- Aftershock sequences (Poisson mainshock occurrence only — SEIS-L2).
- Liquefaction or landslide as independent primary hazards.
- Business-interruption quantum (the DS3 leg is fixed at 100% loss).
- Explicit floor-level geometry (height enters only through `NumberOfStoreys`).

3 Mathematical Framework

3.1 Occurrence (Model A)

Earthquake occurrence is a homogeneous Poisson process. The per-asset annual rate is

$$\lambda_i = \lambda_{\text{zone}}\, m_{\text{fault}}\, m_{\text{soil}},$$

where λ_{zone} is the mean annual rate of events $M \geq m_{\min}$ at the asset's seismic hazard zone (Appendix A), $m_{\text{fault}} = 1.0 + \delta_{\text{GS01}}\, \mathbf{1}[\text{FaultProximityKm} < 1.0 \text{ and GS01 absent}]$ with $\delta_{\text{GS01}} = 0.15$, and m_{soil} is the soil recurrence modifier. Over a run horizon h, $N \sim \text{Poisson}(\lambda_i h)$ is sampled per draw; an earthquake instantiates when $N \geq 1$. The magnitude is drawn from the truncated Gutenberg-Richter law

$$P(M > m) = \frac{10^{-bm} - 10^{-b\,m_{\max}}}{10^{-b\,m_{\min}} - 10^{-b\,m_{\max}}},$$

with $b = 1.0$, $m_{\min} = 5.0$, $m_{\max} = 7.5$ by default (zone-configurable).

3.2 Fault Geometry and Spatial Draw

For each instantiated earthquake the rupture location is sampled from the catchment fault-trace polyline. Cumulative arc-length $S(k)$ is computed at each vertex with the Haversine formula; a uniform draw u selects $s^\star = u\,S(n)$ and the rupture centroid is linearly interpolated. The surface rupture length follows the Wells-Coppersmith scaling $\log_{10} L_r = -2.44 + 0.59\,M$, and the rupture occupies $[s^\star - L_r/2,\, s^\star + L_r/2]$ clipped to the fault endpoints. The Joyner-Boore distance R_{JB} is the minimum distance from the asset to the rupture segment — the same flat-Earth polyline primitive the flood module uses for river-bank distance, so no new GIS dependency is introduced.

3.3 Ground Motion Intensity (Model B)

The site PGA is sampled from a GMPE in lognormal form,

$$\ln(\mathrm{PGA}) = \ln \overline{\mathrm{PGA}}(M, R_{JB}, V_{S30}) + \sigma_{\mathrm{tot}}\,\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0,1),$$

with $\sigma_{\mathrm{tot}} = 0.65$ by default. Soil amplification enters implicitly through V_{S30} (mapped from `SoilType` or read directly from `SoilVs30Mps`). When GS02 is absent and the site class is D or E an additional factor $F_a = 1.30$ multiplies the GMPE median before sampling. The seed GMPE is a parametric Boore-Atkinson-style form per tectonic regime, calibrated to reproduce the indicative attenuation of Table 3.

3.4 Seismic Response Effectiveness (Model C)

Model C is deterministic. The fragility median multiplier per damage state is the product of upward shifts for each present measure,

$$\theta_i^{\mathrm{adj}} = \theta_i^{\mathrm{base}}\,\theta_i^{\mathrm{mult}}, \qquad \theta_i^{\mathrm{mult}} = \prod_j \bigl(1 + \delta_j\,\mathbf{1}[\text{measure } j \text{ present}]\bigr),$$

with the modifiers δ_j of Appendix C. The governance measures GS16–GS18 contribute a single combined factor capped at +0.05. Model C also emits the recovery acceleration r_{acc} (from the BRI geoseismic rating, with a GS18 uplift) and the post-earthquake-fire probability p_{PEF} (reduced 80% by GS15).

3.5 Damage State, Loss and Resilience Credit (Model D)

For each instantiated earthquake the lognormal fragility CDF is evaluated at the sampled PGA for each damage-state threshold,

$$P(\mathrm{DS} \geq \mathrm{DS}_i \mid \mathrm{PGA}) = \Phi\left(\frac{\ln(\mathrm{PGA}) - \ln(\theta_i^{\mathrm{adj}})}{\beta}\right), \qquad \beta = 0.60,$$

and the highest DS_i with $P(\mathrm{DS} \geq \mathrm{DS}_i) > U$ is assigned ($U \sim \mathrm{Uniform}(0,1)$). A loss draw conditioned on the damage state gives the structural loss ratio (DS3 fixed at 1.0). If DS $\geq$ DS2 a post-earthquake-fire cascade may fire ($V < p_{\mathrm{PEF}}$); the fire model is invoked with `InitiationClass = ExternalExposure` and the scenario loss becomes $\max(L_{\mathrm{seismic}}, L_{\mathrm{fire}})$. A functionality-recovery curve $Q(t)$ ramps from $Q(t_0)$ to 1.0 over $T_{\mathrm{rec}}/r_{\mathrm{acc}}$.

Aggregating over n_{sim} draws (of which n_{events} instantiate earthquakes),

$$\text{loss frequency} = \frac{\sum_s L_s}{n_{\mathrm{sim}}}, \quad \text{no-collapse rate} = \frac{n_{\mathrm{DS0}} + n_{\mathrm{DS1}} + n_{\mathrm{DS2}}}{n_{\mathrm{events}}}, \quad \text{seismic spread (bps)} = \frac{n_{\mathrm{DS3}}}{n_{\mathrm{sim}}} \times 10{,}000.$$

The seismic spread is combined with the flood, wind and fire legs by root-sum-of-squares: all-in = $\sqrt{\sum_{\mathrm{perils}} \mathrm{leg}^2}$.

4 Input Parameters and Calibration

All numeric seeds live in `config/seismicmatrices.json` and `config/seismic_zones.json` (`version "seed-0"`) and are loaded by `config/seismic`; no number is embedded in the model code. The resilience-measure vocabulary (GS01–GS18) is single-sourced from the CDM via the model's CDM adapter. *Calibration status:* every value is a first-pass engineering-judgement seed chosen for plausible direction and ordering, not magnitude. Statistical calibration against historical commercial seismic-loss data (SEIS-R1) and independent GMPE benchmarking (SEIS-R2) are open remediation items; until they close the outputs must be read as relative resilience signals, not absolute loss probabilities. The full parameter set is reproduced in the Appendices.

5 Implementation Details

5.1 Module Layout

- `config/seismic/` — parameter schema (enums, dataclasses, loader); `seismicmatrices.json` and `seismic_zones.json` — seed numeric parameters.
- `src/models/seismic/datastructures/` — runtime value types.
- `src/models/seismic/occurrence/` — Model A (Poisson draw, magnitude sample, fault spatial draw, R_{JB}).
- `src/models/seismic/groundmotion.py` — Model B (GMPE).
- `src/models/seismic/responseeffectiveness.py` — Model C.

- `src/models/seismic/damage.py` — Model D, plus the `simulate_asset_seismic` / `simulate_portfolio_se` orchestrators.
- `app/commands/port/stages/seismic.py` — the port stage writing `seismic/seismic.json`.
- `src/routes/commercial/hazard/` — the read-time join attaching `seismic_spread_bps` to the asset hazard payload.

5.2 Randomness

All randomness flows through a single caller-owned `numpy.random.Generator`, so an entire portfolio run is reproducible from one seed — matching the storm, typhoon and fire convention. Model A draws the Poisson count, the magnitude and the rupture location; Model B draws one GMPE residual per event; Model D draws the damage-state uniform, the loss sample and the cascade gate. Model C is deterministic.

6 Validation and Backtesting

Historical calibration and backtesting are open remediation items (SEIS-R1, SEIS-R2) to be completed before production sign-off. Current validation is by directional unit tests, which confirm the model behaves as designed: a BRI-AA asset has materially lower DS3 frequency than a BRI-NR asset; base isolation (GS12) produces the largest single-measure shift; low-hazard zones have near-zero collapse frequency; the post-earthquake-fire cascade fires at the expected conditional rate; and a whole-portfolio run is bit-for-bit reproducible from a single seed.

6.1 Automated Test Results

The model is exercised by 87 unit tests across `tests/models/seismic/` plus route-join and loan-coupon integration tests, and the sensitivity harness asserts the six Section 7 acceptance gates.

Table 1: Test suites covering MKM-SEIS-001

Suite	Scope	Tests
`test_config`	Config schema + seed values	18
`test_cdm_adapter`	CDM $\rightarrow$ features mapping	12
`test_occurrence`	Model A (occurrence + fault draw)	20
`test_groundmotion`	Model B (GMPE)	14
`test_responseeffectiveness`	Model C (BRI modifiers)	11
`test_damage`	Model D (fragility, loss, orchestrator)	17
`test_commercial_routes_part5`	PRS-pricer read-time join	5
`test_loan_pricer_standalone_part4`	Loan-Calculator RSS coupon	6

Table 2: Section 7 acceptance gates (n_{sim} = 20,000 per cell)

Gate	**Result**
Distance monotonic (spread non-increasing in R_{JB})	PASS
Intensity monotonic (spread non-decreasing in zone)	PASS
Resilience: no-collapse non-decreasing NR $\rightarrow$ AA	PASS
Resilience: spread non-increasing NR $\rightarrow$ AA	PASS
Interaction (Very high, NR) $>$ (High, AA) $\geq$ (Moderate, AA)	PASS
Near-zero floor (Very low, R_{JB} = 100 km) $<$ 1 bps	PASS

7 Sensitivity Analysis

The tables below are generated by `docs/models/sensitivities/seismic_resilience` running the real `simulate_asset_seismic` against the baseline of Section 7.1 and varying one input at a time over 20,000 Monte Carlo draws per cell. The baseline asset is a 4-storey Office in Fair condition, RC pre-code construction, stiff soil (class B), Moderate seismic zone, located 20 km from the fault trace, with all BRI geoseismic measures absent.

The model is required to exhibit monotonic and material sensitivity along each of the three primary axes — distance, intensity and resilience — and to satisfy five acceptance conditions: (i) distance monotonicity (spread non-increasing in R_{JB}); (ii) intensity monotonicity (spread non-decreasing in zone); (iii) resilience monotonicity (no-collapse non-decreasing and spread non-increasing from NR to AA); (iv) interaction ordering $(\text{Very high}, \text{NR}) > (\text{High}, \text{AA}) \geq (\text{Moderate}, \text{AA})$; and (v) a near-zero floor ($<$ 1 bps in the Very low zone at R_{JB} = 100 km). All five gates pass against the current seeds.

Table 3: Seismic credit by source-to-site distance R_{JB} (RC pre-code, Moderate zone, all BRI absent)

R_{JB}	**Median PGA (g)**	**No-collapse (%)**	**Loss freq (%)**	**Spread (bps)**
2 km	0.1157	93.1	0.1228	5.00
5 km	0.0835	94.4	0.0954	4.00
10 km	0.0520	97.9	0.0526	1.50
20 km	0.0291	100.0	0.0151	0.0000
50 km	0.0129	100.0	0.0088	0.0000
100 km	0.0068	100.0	0.0083	0.0000

Table 4: Seismic spread (bps) by R_{JB} × construction type (Moderate zone, all BRI absent)

R_{JB}	**RC pre-code**	**Masonry URM**	**RC modern**	**Steel MRF**
2 km	5.00	9.50	3.50	2.50
10 km	1.50	3.00	0.0000	0.0000
20 km	0.0000	0.5000	0.0000	0.0000
50 km	0.0000	0.0000	0.0000	0.0000

Table 5: Seismic credit by hazard zone (RC pre-code, $R_{JB} = 20$ km, all BRI absent)

Zone	No-collapse (%)	Loss freq (%)	Spread (bps)
Very low	100.0	0.0012	0.0000
Low	100.0	0.0061	0.0000
Moderate	100.0	0.0151	0.0000
High	99.1	0.1067	2.50
Very high	99.4	0.2868	4.50

Table 6: Seismic spread (bps) by hazard zone × soil type (RC pre-code, $R_{JB} = 20$ km) — the soft-soil amplification and GS02 gate

Zone	Rock (A)	Stiff (B)	Soft (D) no GS02	Soft (D) GS02
Low	0.0000	0.0000	0.0000	0.5000
Moderate	0.0000	0.0000	2.50	2.50
High	2.00	2.50	9.50	4.00
Very high	5.00	4.50	33.0	21.5

Table 7: Seismic credit by BRI resilience level (RC pre-code, Moderate zone, near-fault $R_{JB} = 5$ km)

BRI level	No-collapse (%)	Loss freq (%)	Spread (bps)
NR	94.4	0.0954	4.00
B	97.2	0.0645	2.00
A	99.3	0.0445	0.5000
AA	100.0	0.0226	0.0000

Table 8: Seismic credit by individual BRI measure (single-measure sweeps; Moderate zone, near-fault $R_{JB} = 5$ km, RC pre-code, stiff soil)

Measure	No-collapse (%)	Spread (bps)	Δ spread
None (baseline)	94.4	4.00	0.0000
GS08: Lateral force resistance	95.8	3.00	-1.00
GS09: Connected / braced	95.8	3.00	-1.00
GS12: Base isolation	97.2	2.00	-2.00
GS14: No vertical irregularities	95.1	3.50	-0.5000
GS02: Seismic foundation (F_a gate)	94.4	4.00	0.0000

Table 9: Seismic spread (bps) by hazard zone × BRI resilience level (RC pre-code, $R_{JB} = 20$ km) — strong measures are most valuable in high-hazard zones

Zone	NR	B	A	AA
Very low	0.0000	0.0000	0.0000	0.0000
Low	0.0000	0.0000	0.0000	0.0000
Moderate	0.0000	0.0000	0.0000	0.0000
High	2.50	2.00	2.00	0.0000
Very high	4.50	6.00	3.50	1.00

Table 10: Seismic spread (bps) by R_{JB} × BRI resilience level (RC pre-code, Moderate zone)

R_{JB}	**NR**	**B**	**A**	**AA**
2 km	5.00	4.00	2.00	0.5000
10 km	1.50	0.5000	0.0000	0.0000
20 km	0.0000	0.0000	0.0000	0.0000
50 km	0.0000	0.0000	0.0000	0.0000

7.1 Portfolio resilience credit — worked example

The tables above vary one input from a synthetic baseline. To show the credit on a real book, Table 11 prices an actual 20-asset commercial portfolio (the Halong catchment, RC pre-code / modern construction, near the Red River Fault, in the High and Very-high hazard zones) at each BRI geoseismic resilience level. The assets, fault geometry and random seed are held fixed; only the installed measures change.

Table 11: Seismic spread by BRI resilience level — 20-asset Halong commercial portfolio (20,000 draws per asset, fixed seed). The credit is the spread reduction relative to the un-resilient (NR) configuration.

BRI level	**Mean spread (bps)**	**Max (bps)**	**No-collapse (%)**	**Credit vs NR**
NR (no measures)	14.57	101.5	95.89	—
B (basic)	10.32	71.5	96.59	−29%
A (strong)	7.10	39.5	97.92	−51%
AA (base-isolated)	3.30	19.5	99.06	−77%

A portfolio with no seismic resilience measures prices at ∼14.6 bps; the same portfolio with best-practice measures (lateral-force resistance, bracing, connections and seismic base isolation) prices at ∼3.3 bps — a 77% reduction, with the worst single-asset spread falling from 101.5 to 19.5 bps and the no-collapse rate rising from 95.9% to 99.1%. The NR→B→A steps are roughly linear; the largest single jump is A→AA (−54%), driven by GS12 seismic base isolation (a +0.40 shift on every fragility median) — the headline non-linearity the credit is designed to price.

8 Model Limitations and Known Weaknesses

- **No historical calibration (SEIS-L1).** Every parameter is a first-pass placeholder; outputs are relative signals, not absolute loss probabilities.
- **Mainshock-only Poisson (SEIS-L2).** The HPP assumption ignores aftershock clustering; appropriate for long-run annualised loss but not real-time post-event risk.
- **Single fault trace per catchment (SEIS-L3).** Multiple independent sources require a superposition of Poisson processes (planned, SEIS-R4).
- **Deterministic GMPE median (SEIS-L4).** Only the aleatory residual is sampled; epistemic logic-tree uncertainty is not represented.
- **Height proxy (SEIS-L5).** Building height enters only through `NumberOfStoreys`.

9 Governance Validation Questions

The following nine questions are mandated by Chapter 5 of the *Model Risk Governance for Vendors* handbook and must be explicitly addressed for every model in the inventory. Response status is tracked in the Model Governance Dashboard (MRC portal).

9.1 Q1 — Model purpose and use

"The model produces a seismic-resilience credit (loss frequency, no-collapse rate and a full-collapse bps leg) consumed by the Commercial PRS Pricer as an independent peril. Use is consistent with design; it is not a life-safety or absolute-loss model and must not be used as one."

9.2 Q2 — Data quality and lineage

"Inputs are the CDM-derived `AssetSeismicFeatures`*; lineage runs from the v2 CDM resilience fields (GS01–GS18) through Model C. Seismic zone and fault-trace files are catchment-level spatial assets versioned alongside the model. Missing GS flags default to the absence case. Calibration data quality is not yet established (SEIS-R1)."*

9.3 Q3 — Conceptual soundness

"The four-model decomposition (Poisson occurrence with a fault-geometry spatial draw, GMPE intensity, BRI measure mapping, lognormal fragility with a loss draw) is the standard PSHA/PBEE framework used in HAZUS, GEM and Eurocode 8 risk assessment. The Poisson mainshock assumption is well supported for engineering annual-loss estimation."

9.4 Q4 — Calibration and parameters

"All parameters are engineering-judgement seeds (`seed-0`*) chosen for direction and ordering, not magnitude. Statistical calibration is open (SEIS-R1, SEIS-R2); this is the dominant model limitation."*

9.5 Q5 — Implementation verification

"Verified by unit tests covering each component model, the orchestrator and the cascade gate, plus a reproducibility test. The sensitivity harness exercises the full path and passes the five Section 7 acceptance gates."

9.6 Q6 — Sensitivity and stability

"See Section 7. The model exhibits monotonic and material sensitivity along distance, intensity and resilience, with the designed intensity × resilience non-linearity (strong BRI measures most valuable in high-hazard zones). Stability is adequate at $n_{sim} = 20{,}000$."

9.7 Q7 — Limitations and weaknesses

"Documented in Section 8 (SEIS-L1–L5). The placeholder calibration and the single fault-trace restriction are the most material."

9.8 Q8 — Ongoing monitoring

"Not yet in production. A monitoring plan (input-distribution drift; backtest of realised vs predicted seismic-loss frequency) is defined under SEIS-R3 before sign-off."

9.9 Q9 — Governance and approval

"Owner: David K Kelly. Status: Development, Tier 2, RAG Amber. Not approved for production; MRC sign-off is contingent on closing SEIS-R1–R5."

10 Change History

Version	Date	Description
0.10	08-June-2026	Stages 0–9: Models A–D, port stage, PRS-pricer and Loan-Calculator integration, se

A Occurrence Parameters (Model A)

Table 12: Seismic hazard zone annual rates ($M \geq 5.0$ mainshock) — seed-0

Zone	λ_{zone} (/yr)	b	$m_{\max}$	Notes
Very low	0.0005	1.0	6.5	Intraplate stable continental
Low	0.002	1.0	7.0	e.g. UK, NW Europe
Moderate	0.008	1.0	7.5	e.g. central Italy, Romania
High	0.030	1.0	8.0	e.g. Turkey, Greece, coastal Vietnam
Very high	0.080	1.0	8.5	e.g. Japan, Philippines, Sumatra

Table 13: Fault proximity (GS01) and soil recurrence modifiers

Condition	Modifier
FaultProximityKm < 1.0 and GS01 absent	$m_{\text{fault}} = 1.15$
Otherwise	$m_{\text{fault}} = 1.00$
Soil class A / B	$m_{\text{soil}} = 1.00$
Soil class C	$m_{\text{soil}} = 1.10$
Soil class D	$m_{\text{soil}} = 1.15$
Soil class E	$m_{\text{soil}} = 1.20$

B GMPE and Site Amplification (Model B)

Table 14: Site-class V_{S30} proxies and the GS02 amplification gate

SoilType	EC8	V_{S30} (m/s)	F_a if GS02 absent
Rock	A	800	1.00
Stiff soil	B	450	1.00
Dense soil	C	270	1.00
Soft soil	D	150	1.30
Very soft	E	100	1.30

The default GMPE is a Boore-Atkinson 2014 (NGA-West2) parametric proxy for active shallow crust ($\sigma_{\text{tot}} = 0.65$) and a Zhao et al. 2006 proxy for subduction interface ($\sigma_{\text{tot}} = 0.68$), selectable per catchment zone.

C Response Effectiveness Modifiers (Model C)

Table 15: Fragility median multiplier δ_j by BRI geoseismic measure — seed-0

Measure	Applies to	δ_j
GS03 Foundation piling to rock	DS2, DS3	+0.12
GS04 Subsidence foundation design	DS2, DS3	+0.08
GS08 Lateral force resistance	DS2, DS3	+0.20
GS09 Connected and braced structure	DS1–DS3	+0.12
GS10 Steel reinforced walls	DS1, DS2	+0.08
GS12 Seismic base isolation	All DS	+0.40
GS14 No vertical irregularities	DS2, DS3	+0.15
GS16–GS18 Governance (combined cap)	All DS	+0.05
GS02 Seismic foundation (soil gate)	Removes F_a (class D/E)	—
GS15 Flexible gas piping / shut-off	p_{PEF}	−80%

Table 16: Recovery acceleration r_{acc} by BRI geoseismic rating

Rating	r_{acc}
AA	1.00
A	0.67
B	0.40
NR	0.25

D Fragility and Damage-State Parameters (Model D)

Table 17: Baseline fragility median PGA (g) by construction class — seed-0

Construction class	**DS1**	**DS2**	**DS3**	β
RC modern seismic	0.15	0.35	0.65	0.60
RC pre-code	0.10	0.22	0.42	0.60
Masonry reinforced	0.12	0.28	0.52	0.60
Masonry unreinforced	0.07	0.14	0.28	0.60
Steel moment frame	0.18	0.40	0.75	0.60
Timber frame	0.12	0.28	0.55	0.60
Default	0.10	0.22	0.42	0.60

Table 18: Loss, functionality and post-earthquake-fire parameters — seed-0

Damage state	**Mean** L	$Q(t_0)$	$T_{\mathbf{rec}}$ **(mo)**	$p_{\mathbf{PEF}}$ **(no GS15)**
DS0 None/Slight	0.01	1.00	0	0.00
DS1 Moderate	0.10	0.75	3	0.00
DS2 Extensive	0.35	0.30	12	0.08
DS3 Complete	1.00	0.00	36	0.18

DS3 loss is fixed at 1.0 (no draw); DS0–DS2 losses are truncated-lognormal on $[0, 1]$ with the means above and log-standard deviations 0.50/0.50/0.40. GS15 reduces p_{PEF} by 80%. T_{rec} is divided by r_{acc} to give the effective recovery time.